

nology: The Inside Story

One inherent shortcoming with DSTN displays is the fact that the response of the elements is very slow. This results in a smearing caused in the image due to the inability of the screen to update fast enough to display new signals. The older DSTN displays use a material that typically has a response time of about 300 ms—almost a third of a second—making these displays highly inappropriate for fast moving graphics and multimedia-intensive applications.

One of the most prevalent technologies in the LCD arena that has proved to be very effective in both speed and image contrast is TFT. With response times of 25 ms, these displays are quite suitable for multimedia and video applications that need the screen to have fast switchover times.

TFT screens are constructed using an additional transistor to drive each of the RGB lines for every pixel. Since the switching capability of these transistors is very fast, the problems of ghosting and blurring of the images are significantly reduced. One of the disadvantages with TFT screens is that if any of the transistors fail, that pixel on the screen will remain permanently off. In its infancy, LCD screens used to have the crystalline material placed such that it ran vertically from the top to the bottom of the screen with the voltages changing the direction of these crystals and permitting light to pass through. Therefore, the viewing angle of these screens was rather restricted.

TFT screens today use a crystalline material that is naturally aligned in a horizontal plane, allowing for viewing angles of up to 140 degrees—a very significant

In the review

LCD technology in displays has been around for a while. Laptops have widely used LCD displays over the last decade or so due to the compactness of the screens and the lower power consumption when compared to CRT-based displays.

While CRT monitors have distinct advantages such as good colour reproduction, support for a wider range of resolutions, smoother image reproduction and a wide range of viewing angles, LCD monitors do offer certain advantages. Apart from the compactness, which is a distinct plus point where space is an issue, you also experience much less eye strain with LCD monitors.

Yet, LCDs have not really been in the reckoning because of the high prices and relatively inefficient manufacturing processes.

However, with recent advancements in technology, several of these issues have been taken care of and over the last

two or three years, the number of LCD displays available for desktops has been steadily increasing. This, along with the dramatic reduction in prices over the last year (a drop of almost 25 per cent), makes LCD monitors an increasingly attractive alternative.

Our Test Centre received models from manufacturers such as Samsung, Sharp, Acer, Convergent and Philips.

However, considering that at Rs 45-50,000 most of these displays cost as much as some complete systems, they are still just out of the reach of most PC buyers. For this reason the eight LCD models we received have been reviewed in the course of this test for the features offered but no rating or comparison was conducted. Although no ratings were awarded and no winner was decided, the reviews on the CD are meant to give you a fair idea of whether a particular product is the right one for you.



increase over that of passive matrix' 40 to 100 degrees.

The only present disadvantage with TFTs is the pixel size. Until now, manufacturers have not been able to get below the 0.3 mm limit, which is way too large by monitor standards. Therefore, when this technology is adapted to 15- and 17-inch computer displays the pixellation, when compared to a high-end monitor is significantly more apparent.

The newest range of third-generation TFT screens designed specifically for the desktop also feature the capability of being driven through digital outputs from the newer graphics cards. This connector, known as DFP (Digital Flat Panel) connec-

tor, eliminates the need to convert an analog video signal to digital format for LCD displays. Staying in the digital realm, there is greater noise immunity and therefore greater image clarity with this method.



More Info on

Test process and reviews

Acer FP563

Convergent CT170

Convergent XTAL6220

Philips 150X

Philips Brilliance 150P

Samsung SyncMaster 151MP

Samsung SyncMaster 570S

Sharp LL — T1501A

check out

Mindware